

DHRUV BHAGAT

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SUMMARY

Computer Science undergraduate with experience in software engineering, backend development, Machine Learning, and AI-powered application development. Skilled in building scalable applications, RESTful APIs, and intelligent software solutions using Python, PyTorch, FastAPI, and Node.js. Strong foundation in OOP, Data Structures & Algorithms, SDLC, databases, and cloud technologies, with a passion for developing reliable and impactful software.

EXPERIENCE

AI for Sustainability Intern | 1M1B, AICTE & IBM SkillsBuild May 2026 – Present

- Developed EcoSort AI, an end-to-end waste classification system using PyTorch, EfficientNet-B0, FastAPI, Node.js, and MongoDB for real-time waste recognition and intelligent disposal recommendations.
- Trained and optimized deep learning models on 27,000+ images across 10 waste categories, achieving ~94% classification accuracy using transfer learning and model evaluation techniques.

Mobile App Developer & Game Developer May 2025 – June 2025 *Intern at TejasKP AI Software* Vadodara, India

- Engineered and deployed 'Kaushal Associates of Lubrication', a Flutter-based mobile application supporting 2 user roles with vehicle service booking, real-time request tracking, and workflow management.
- Designed 5 interactive 3D vehicles for a Unity-based car racing game, implementing realistic vehicle physics, collision detection, scoring systems, and gameplay mechanics.

EDUCATION

Chandubhai S. Patel Institute of Technology (CSPIT), Charusat University 8.71/10 *Bachelor of Technology (B.Tech) – Computer Science and Engineering*

PROJECTS

EcoSort AI – Smart Waste Classification System [GitHub](#)

- Created an AI-powered waste classification system using EfficientNet-B0 and PyTorch for intelligent waste segregation.
- Trained and fine-tuned the model on 27,000+ images across 10 waste categories, achieving ~94% classification accuracy.
- Designed and implemented 11 REST API endpoints (9 Express.js, 2 FastAPI) to enable real-time AI inference, waste analytics, eco-score generation, and intelligent disposal recommendations.

Safeguard – Smart IoT Suicide-Prevention & Emergency Response System [GitHub](#)

- Built a Flutter-based IoT mobile application integrating ESP32 and Firebase for real-time emergency monitoring and response.
- Developed role-based authentication and automated alert workflows for students, parents, wardens, and administrators to improve emergency communication.
- Integrated real-time sensor data with the mobile application, enabling event detection, incident tracking, and push notifications.

End to End E-Commerce Churn Intelligence System [GitHub](#)

- Created and deployed a cloud-based churn prediction system using Random Forest, achieving 93.82% ROC-AUC.
- Built REST APIs for real-time prediction services and automated workflows using n8n.
- Containerized the application using Docker and deployed it on Render and Vercel for production-ready performance.

TECHNICAL SKILLS

- Languages:** Python, JavaScript, SQL
- Frontend:** React.js, TypeScript
- Backend & APIs:** Node.js, Express.js, Django, FastAPI, RESTful APIs
- Machine Learning & AI:** PyTorch, Deep Learning, Computer Vision, Transfer Learning, LLMs, Generative AI, Prompt Engineering
- Databases:** MongoDB, PostgreSQL
- Cloud & DevOps:** AWS (EC2, S3, Lambda, API Gateway, DynamoDB), Docker, Git, CI/CD (Basics)
- Software Engineering:** OOP, Data Structures & Algorithms, SDLC, STLC, Agile Methodologies
- Testing & Automation:** PyTest (Basics), Appium (Basics), n8n Workflow Automation, Postman

CERTIFICATIONS

- IBM – Machine Learning with Python
- AWS Academy Graduate – Cloud Developing
- Microsoft – Foundations of AI and Machine Learning
- NPTEL – Design and Analysis of Algorithms